

2627-MA140: Week 03, Lecture 1 (L07)

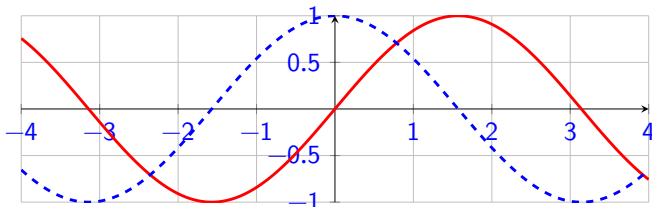
2627-MA140: Week 03, Lecture 1 (L07) The Squeeze Theorem & one-sided limits

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University of Galway

Tuesday, 22 September, 2026

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Reminder

- ▶ **Assignment 1** has a deadline of 5pm, Monday 28 September. You can access it on Canvas... 2628-MA140... Assignments... **Problem Set 1**. Click on **Problem 1** (e.g.) to get started.
- ▶ The **Tutorial Sheet** is available at <https://universityofgalway.instructure.com/courses/58477/files/3810601>
- ▶ Assignment 2 will open by Friday.

The first of the two class tests will take place 2 weeks from now:

Tuesday, 6th October.

- ▶ You will have 40 minutes to complete the test, which will be composed of multiple choice questions.
- ▶ Main venues: ENG-G017 or ENG-G018 (you'll be told which)
- ▶ I need to gather information on Reasonable Accommodations for tests. If you want to avail of such, please complete this form: <https://forms.cloud.microsoft/e/rBiq8RHYY2> by 10am, Tuesday 29 September.

2627-MA140 - Class Test
Accommodations



Topics we'll squeeze into today's class:

1 The Squeeze Theorem

2 $\sin(\theta)/\theta$

- Other examples

3 Infinite Limits

4 Digression: How fast can an object travel?

5 One-sided Limits

- Notation
- Piecewise functions

- Empty and full circle notation

- Existence of a limit

6 Exercises

For more, see Chapter 2 (Limits) of Strang and Herman's **Calculus**, especially Sections 2.3 (Limit Laws) and 2.2 (Limit of a Function).

The Squeeze Theorem

Last week we learned about the formal definition of a limit, and the “Limit Laws” for calculating the limit of certain functions, such as rational functions.

We'll now study another method for computing the limit of a (complicated) function, by “trapping” it between two simpler functions.

$$\text{Eg: } \lim_{x \rightarrow 0} \frac{\sin(x)}{x}$$

The Squeeze Theorem

The Squeeze Theorem (a.k.a. Sandwich Theorem)

Suppose that we have three functions f , g and h on some interval $[x_0, x_1]$, with

$$\underline{g(x)} \leq \underline{f(x)} \leq \underline{h(x)},$$

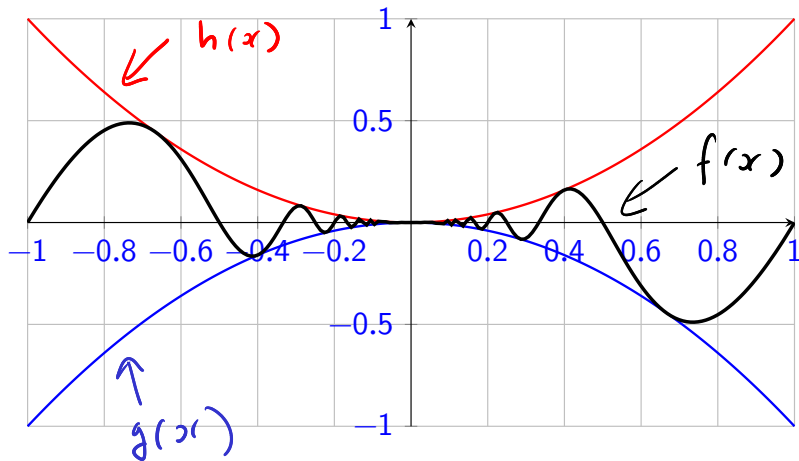
and

$$\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} h(x) = L,$$

for some $a \in [x_0, x_1]$. Then $\lim_{x \rightarrow a} f(x) = L$.

That is: if $g(x)$ and $h(x)$ have the same limit as $x \rightarrow a$, and $f(x)$ is “squeezed” between them, then so too does $f(x)$.

The Squeeze Theorem



Example

Suppose $f(x)$ is a function such that

$$1 - \frac{x^2}{4} \leq f(x) \leq 1 + \frac{x^2}{2}, \quad \forall x \neq 0$$

Find $\lim_{x \rightarrow 0} f(x)$.

"for all x except $x=0$ "

Check: $g(x) = 1 - x^2/4$ $h(x) = 1 + x^2/2$

$$\lim_{x \rightarrow 0} g(x) = 1 \quad \& \quad \lim_{x \rightarrow 0} h(x) = 1$$

So, since $g(x) \leq f(x) \leq h(x) \Rightarrow \lim_{x \rightarrow 0} f(x) = 1$

$$\sin(\theta)/\theta$$

Next, we will use the Squeeze Theorem to explain **an important limit**:

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

First: check that we can't just evaluate $f(\theta) = \frac{\sin(\theta)}{\theta}$ at $\theta = 0$.

(Note $\sin \theta$ is really $\sin(\theta)$.)

$$\sin(0) = 0$$

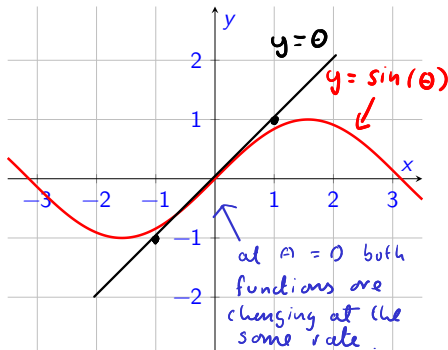
$$\text{So at } \theta = 0, \frac{\sin(\theta)}{\theta} = \frac{0}{0}$$

So $\theta = 0$ is not in the domain of $\frac{\sin(\theta)}{\theta}$

$$\sin(\theta)/\theta$$

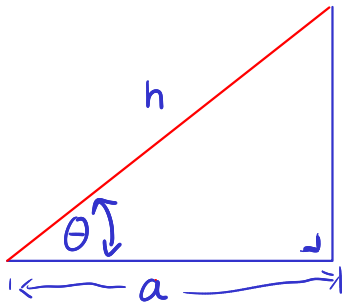
Next, First, let's just convince ourselves that it is likely that

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1.$$



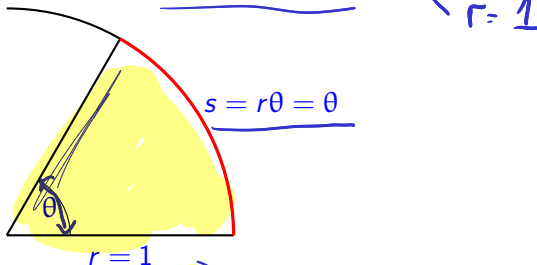
Next, we review some facts about trigonometric functions.

- ▶ **In this module, we only every use radians** (never degrees).
- ▶ Given a right-angled triangle, with sides a , b and h (the hypotenuse), $\sin \theta = \frac{b}{h}$, $\cos \theta = \frac{a}{h}$, $\tan \theta = \frac{b}{a} = \frac{\sin \theta}{\cos \theta}$



$$\begin{aligned} b \quad \frac{\sin(\theta)}{\cos(\theta)} &= \frac{b}{h} \cdot \frac{h}{a} = \frac{b}{a} \\ &= \tan(\theta). \end{aligned}$$

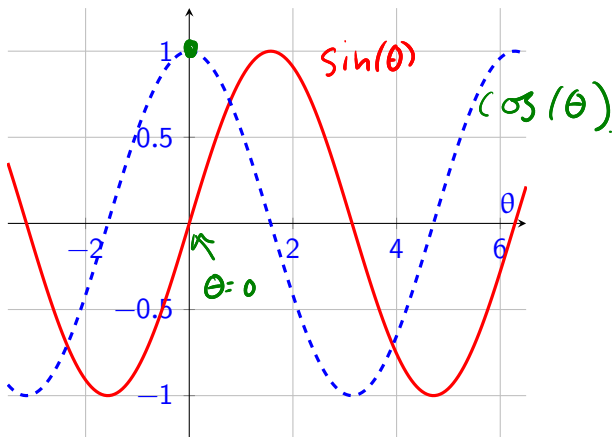
- ▶ The length of a sector, subtended by the angle θ , of a circle of radius r , is $s = r\theta$. In particular, for the unit circle **the angle (in radians) is the length of the arc..**



- ▶ Area of a sector of a circle is $\frac{1}{2}r^2\theta$ where r is the radius of the circle, and θ is the angle subtended by the sector.

$$\sin(\theta)/\theta$$

- The $\sin$ (red) and $\cos$ (blue) functions look like this:



Various other facts are summarised in the State Examination Commission's Tables:

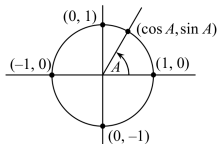
Triantánacht

$$\tan A = \frac{\sin A}{\cos A}$$

$$\sec A = \frac{1}{\cos A}$$

$$\cot A = \frac{\cos A}{\sin A}$$

$$\operatorname{cosec} A = \frac{1}{\sin A}$$



Trigonometry

$$\cos^2 A + \sin^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\cos(-A) = \cos A$$

$$\sin(-A) = -\sin A$$

$$\tan(-A) = -\tan A$$

Nóta: Bíonn $\tan A$ agus $\sec A$ gan sainiú nuair $\cos A = 0$.

Bíonn $\cot A$ agus $\operatorname{cosec} A$ gan sainiú nuair $\sin A = 0$.

Note: $\tan A$ and $\sec A$ are not defined when $\cos A = 0$.

$\cot A$ and $\operatorname{cosec} A$ are not defined when $\sin A = 0$.

A (céimeanna)	0°	90°	180°	270°	30°	45°	60°	A (degrees)
A (raidian)	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	A (radians)
$\cos A$	1	0	-1	0	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	$\cos A$
$\sin A$	0	1	0	-1	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	$\sin A$
$\tan A$	0	-	0	-	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	$\tan A$

$$1 \text{ rad.} \approx 57.296^\circ$$

$$1^\circ \approx 0.01745 \text{ rad.}$$

Foirmlí uillinneacha comhshuite

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

Compound angle formulae

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$\sin(A-B) = \sin A \cos B - \cos A \sin B$$

$$\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

Foirmlí uillinneacha dúbailte

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos^2 A = \frac{1}{2}(1 + \cos 2A)$$

$$\sin^2 A = \frac{1}{2}(1 - \cos 2A)$$

Double angle formulae

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$\cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$$

$$\sin 2A = \frac{2 \tan A}{1 + \tan^2 A}$$

Iolraigh a thiontú ina suimeanna agus ina ndifríochtaí**Products to sums and differences**

$$2 \cos A \cos B = \cos(A + B) + \cos(A - B)$$

$$2 \sin A \cos B = \sin(A + B) + \sin(A - B)$$

$$2 \sin A \sin B = \cos(A - B) - \cos(A + B)$$

$$2 \cos A \sin B = \sin(A + B) - \sin(A - B)$$

Suimeanna agus difríochtaí a thiontú ina n-iolraigh**Sums and differences to products**

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

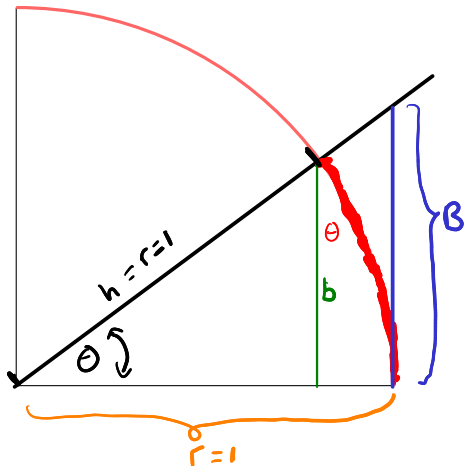
$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\sin(\theta)/\theta$$

Note: We'll use the Squeeze Thm.

To show that $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$, consider the quadrant of the unit circle shown below. $\} r=1$



Note (that $\sin(\theta) = \frac{b}{r} = b$

and $\tan(\theta) = \frac{B}{r} = B$

Therefore $b \leq \theta \leq B$

So $\sin(\theta) \leq \theta \leq \tan(\theta)$.

And, thus,

$$\sin(\theta) \leq \theta \leq \frac{\sin(\theta)}{\cos(\theta)}$$

$$\sin(\theta)/\theta$$

$$\sin(\theta) \leq \theta \leq \frac{\sin(\theta)}{\cos(\theta)}$$

Divide by $\sin(\theta)$ (note $\theta \neq 0$)

$$\Rightarrow \frac{\sin(\theta)}{\sin(\theta)} \leq \frac{\theta}{\sin(\theta)} \leq \frac{\sin(\theta)}{\sin(\theta)\cos(\theta)}$$

$$\Rightarrow 1 \leq \frac{\theta}{\sin(\theta)} \leq \frac{1}{\cos(\theta)}$$

Inverting we get $1 \geq \frac{\sin(\theta)}{\theta} \geq \cos(\theta)$

Applying the Squeeze Theorem to $\cos(\theta) \leq \frac{\sin(\theta)}{\theta} \leq 1$ gives $\lim_{\theta \rightarrow 0} \frac{\sin(\theta)}{\theta} = 1$

where here we have used

that $\cos(0) = 1$, so $\lim_{\theta \rightarrow 0} \cos(\theta) = 1$

and, of course, $\lim_{\theta \rightarrow 0} 1 = 1$.



ExampleEvaluate $\lim_{x \rightarrow 0} \frac{\tan(3x)}{\sin(2x)}$

$$f(x) = \frac{\tan(3x)}{\sin(2x)}$$

Note $\tan(0) = \sin(0) = 0$, so $f(0) = \frac{0}{0}$.

$$\frac{\tan(3x)}{\sin(2x)} = \frac{\sin(3x)}{\cos(3x)} \cdot \frac{1}{\sin(2x)}$$

$$= \frac{\cancel{3x}}{\cancel{2x}} \frac{\sin(3x)}{3x} \cdot \frac{2x}{\sin(2x)} \cdot \frac{1}{\cos(3x)}$$

$$\text{So } \lim_{x \rightarrow 0} f(x) = \frac{3}{2} \lim_{x \rightarrow 0} \frac{\sin(3x)}{3x} \cdot \lim_{x \rightarrow 0} \frac{2x}{\sin(2x)} \cdot \lim_{x \rightarrow 0} \frac{1}{\cos(3x)}$$

Example

Evaluate $\lim_{x \rightarrow 0} \frac{\tan(3x)}{\sin(2x)}$

$$\text{So } \lim_{x \rightarrow 0} f(x) = \frac{3}{2} \cdot (1) \cdot \frac{1}{1} \cdot 1 = \boxed{\frac{3}{2}}$$

Finished here Tuesday

Example

Evaluate $\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta^2}$

Infinite Limits

So far, we've had lots of examples that are a little like:

$$\lim_{x \rightarrow 1} \frac{x^3 - x^2 - x + 1}{(x - 1)^2} = 2.$$

(Check that this is correct).

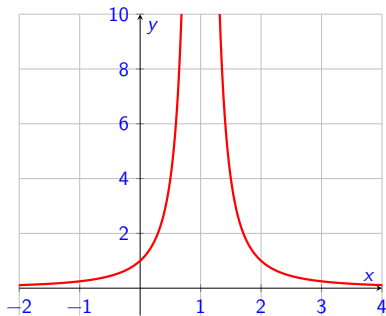
But what about

$$\lim_{x \rightarrow 1} \frac{1}{(x - 1)^2} = ???$$

Let's plot it and see:

Infinite Limits

$$f(x) = \frac{1}{(x-1)^2}$$



As x get closer and closer to 1, the value of $f(x)$ gets larger and larger. In fact, it becomes infinite.

For this we write

$$\lim_{x \rightarrow 1} f(x) = \infty.$$

Digression: How fast can an object travel?

- ▶ Q: Is there any limit to the speed at which an object can travel?
- ▶ A: Yes! (Assuming you believe Einstein)

Thanks to Einstein ($E = mc^2$), Lorenz and others, it is known that the mass of a moving charged particle behaves like

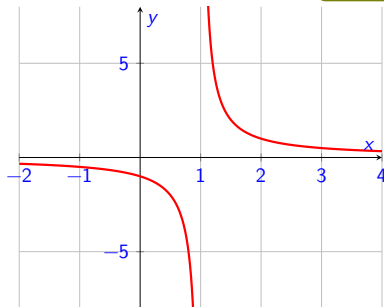
$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

where m_0 is its mass at rest, c is the speed of light, and v is the particles current speed. What happens as $v \rightarrow c$?

One-sided Limits

Let's consider a motivating example, very similar to the one where we introduced ∞ .

$$f(x) = \frac{1}{x-1}$$



As x get closer and closer to 1, then $f(x) \rightarrow -\infty$ or $f(x) \rightarrow \infty$, depending on whether x approaches 1 from the left or right.

To express this, we need the concept of a **one-sided limit**

$\lim_{x \rightarrow a^-} f(x)$ is: **limit of f as x approaches a from the left**

$\lim_{x \rightarrow a^+} f(x)$ is: **limit of f as x approaches a from the right**

In the previous example, with $f(x) = \frac{1}{x-1}$, we have

► $\lim_{x \rightarrow 1^-} f(x) = -\infty$

► $\lim_{x \rightarrow 1^+} f(x) = \infty$

In many important examples, we encounter functions that have different definitions in different regions. The most classic example is the **absolute value function**:

$$|x| = \begin{cases} -x & x < 0 \\ x & x > 0. \end{cases}$$

Care has to be taken when evaluating the limits of such functions....

Example

Sketch the function

$$f(x) = \begin{cases} 3 - x, & x \leq 2 \\ \frac{x}{2} + 1, & x > 2 \end{cases}$$

Find $\lim_{x \rightarrow 2^-} f(x)$ and $\lim_{x \rightarrow 2^+} f(x)$.

Empty and Full Circle Notation:

In the previous sketch, we use the convention that

- ▶ If the end point of a line segment is **not** included in its definition, it terminates with an **open circle**, $\circ$
- ▶ If the end point of a line segment **is** included in its definition, it terminates with an **closed circle** $\bullet$.

Existence of a limit

$\lim_{x \rightarrow a} f(x)$ **exists** if and only if

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$$

So if $\lim_{x \rightarrow a} f(x) = L$ exists, we have

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a^+} f(x) = L$$

though it is not necessary that $f(a) = L$

Example

Sketch the function

$$f(x) = \begin{cases} 3 - x, & x < 2 \\ 4, & x = 2 \\ \frac{x}{2}, & x > 2 \end{cases}$$

Determine if $\lim_{x \rightarrow 2} f(x)$ exists.

Exercise 3.1.1

Suppose that $g(x) = 9x^2 - 3x + 1/4$, and $f(x)$ is such that $-g(x) \leq f(x) \leq g(x)$ for all x .

1. Can one use the Squeeze Theorem to determine $\lim_{x \rightarrow 1/3} f(x)$? If so, do so. If not, explain why.
2. Can one use the Squeeze Theorem to determine $\lim_{x \rightarrow 1/6} f(x)$? If so, do so. If not, explain why.

Exercises

Exercise 3.1.2 (from 2425-MA140 Exam)

Suppose that $g(x) = 2x^4 + x^2$, and $f(x)$ is such that $-g(x) \leq f(x) \leq g(x)$ for all x .

1. Can one use the Squeeze Theorem to determine $\lim_{x \rightarrow 0} f(x)$? If so, do so. If not, explain why.
2. Can one use the Squeeze Theorem to determine $\lim_{x \rightarrow 1} f(x)$? If so, do so. If not, explain why.

Exercise 3.1.3 (from 2023/24 Q1(b))

Evaluate

$$\lim_{\theta \rightarrow 0} \frac{2 \sin(\theta)}{\theta + 3 \tan(\theta)}$$